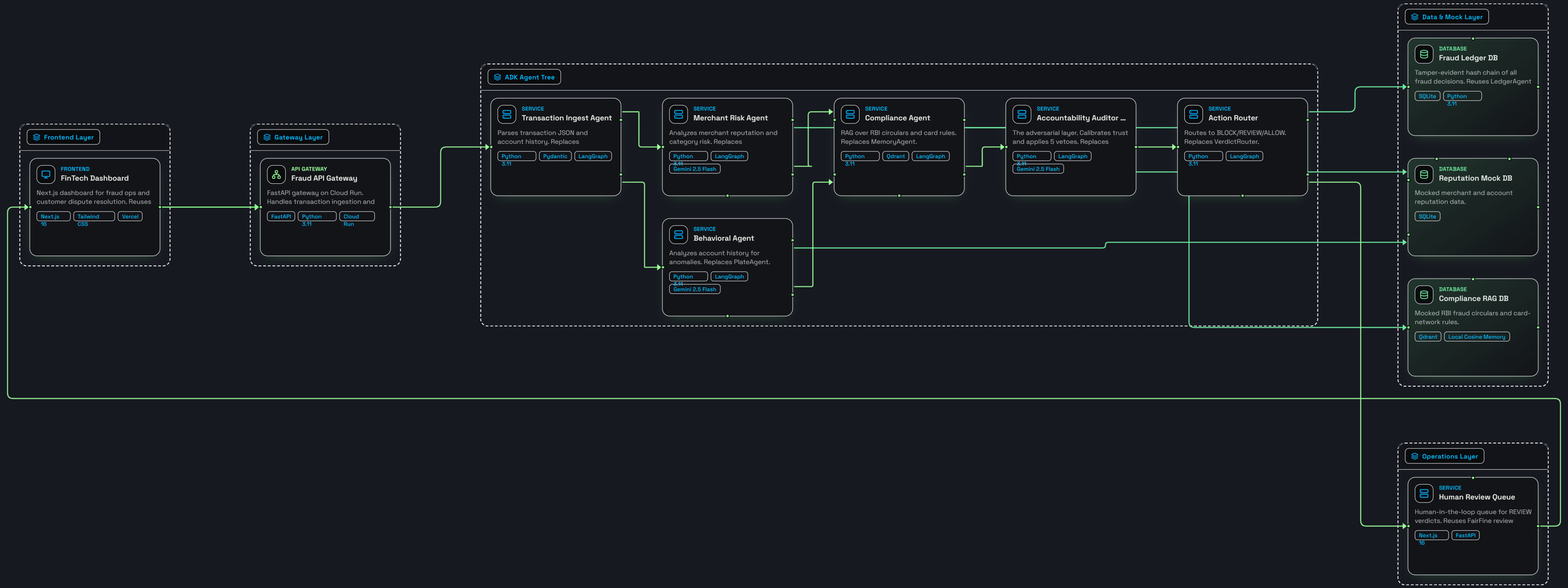
Frontend Layer
FRONTEND
FinTech Dashboard
Next.js dashboard for fraud ops and customer dispute resolution. Reuses
Next.js 16
Tailwind CSS
Vercel
Gateway Layer
API GATEWAY
Fraud API Gateway
FastAPI gateway on Cloud Run. Handles transaction ingestion and
FastAPI
Python 3.11
Cloud Run
ADK Agent Tree
SERVICE
Transaction Ingest Agent
Parses transaction JSON and account history. Replaces
Python 3.11
Pydantic
LangGraph
SERVICE
Merchant Risk Agent
Analyzes merchant reputation and category risk. Replaces
Python 3.11
LangGraph
Gemini 2.5 Flash
SERVICE
Behavioral Agent
Analyzes account history for anomalies. Replaces PlateAgent.
Python 3.11
LangGraph
Gemini 2.5 Flash
SERVICE
Compliance Agent
RAG over RBI circulars and card rules. Replaces MemoryAgent.
Python 3.11
Qdrant
LangGraph
SERVICE
Accountability Auditor ...
The adversarial layer. Calibrates trust and applies 5 vetoes. Replaces
Python 3.11
LangGraph
Gemini 2.5 Flash
SERVICE
Action Router
Routes to BLOCK/REVIEW/ALLOW. Replaces VerdictRouter.
Python 3.11
LangGraph
Data & Mock Layer
DATABASE
Fraud Ledger DB
Tamper-evident hash chain of all fraud decisions. Reuses LedgerAgent
SQLite
Python 3.11
DATABASE
Reputation Mock DB
Mocked merchant and account reputation data.
SQLite
DATABASE
Compliance RAG DB
Mocked RBI fraud circulars and card-network rules.
Qdrant
Local Cosine Memory
Operations Layer
SERVICE
Human Review Queue
Human-in-the-loop queue for REVIEW verdicts. Reuses FairFine review
Next.js 16
FastAPI